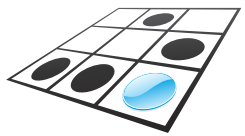
Alt (Sisyphus) on E2K

Total packages: ~ 11500

Core components:

lcc	1.23.20	~ gcc-5.5
Linux kernel	4.9	
glibc	2.23	
gdb	8.1	
strace	4.20	
gtk	2.24	3.24
qt	3 / 4	5.9
boost	1.67	
perl	5.28	
python	2.7	3.7
firefox (+jit)	52.9	
libreoffice	5.4.3.2	



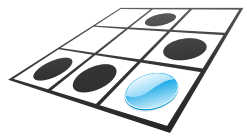
Upstream integration with MCST

- $\Leftarrow$ New versions
- $\Leftrightarrow$ Patches exchange
- $\Rightarrow$ Bug reports

Complexity overview:

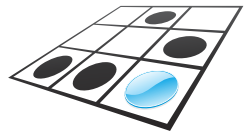
- Hundreds of packages patched for e2k
- Kernel 4.9.170, diff $\sim$ 50 MB:

```
3296 files changed, 1463488 insertions(+),
4390 deletions(-)
```



Free software upstreaming

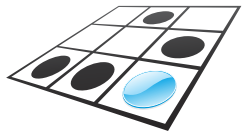
- We can upstream only own patches
  - Sometimes we have mixed patchsets and can’t upstream them
- Upstreams are different:
  - some accept gladly
  - some ignore
  - some demand toolchain support first
- Share of patches fix problems for all architectures detected on e2k



Where e2k is contributed [6]

- binutils*
- bzflag
- gimagereader
- glew
- gnu-config
- gstreamer
- imake
- libtommath
- lxc
- meson*
- netsurf
- double-conversion
- cln6
- nss
- pcre
- ruby
- sanlock
- spice-protocol
- transmission
- util-linux

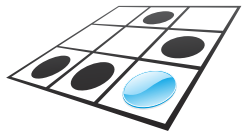
*MCST

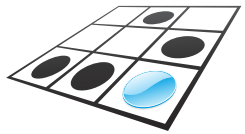


Summary

- Elbrus is self sufficient for a full scale linux distribution
- E2K upstreaming is not easy, but possible
- Awaiting for going public
- Write standard code!

Thank you for your attention!



Bibliography I

- Александр Киирович Ким, Валерий Иванович Перекатов, Сергей Геннадьевич Ермаков. Микропроцессоры и вычислительные комплексы семейства Эльбрус. — Санкт-Петербург : Издательство «Питер», 2013. — ISBN: 978-5-459-01697-0. — http://www.mcst.ru/files/511cea/886487/1a8f40/000000/book_elbrus.pdf.
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